MATH1013 Assignment 1

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LaTeX source available at: codeberg.org/londonwallace/bmath.

1 Exercise 1

Suppose that for each natural number $n \in \mathbb{N}$, A_n is some finite set of real numbers and that A_n and A_m have no members in common if $n \neq m$. **Define**

$$f(x) := \begin{cases} \frac{1}{n} & x \in A_n \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

Compute $\lim_{x\to\infty} f(x)$**.**

Justify with the ϵ and δ definition of a limit.

1.1 Solution

$$\text{The } \lim_{x\to\infty} f(x) = 0 \tag{2}$$

The ϵ and δ definition of a limit reads as follows:

$$\boxed{\text{The function } f \text{ approaches a limit } l \text{ at } a \text{ if: } \forall\epsilon > 0, \exists\delta > 0, \text{ such that if } 0 < |x-a| < \delta, |f(x-l)| < \epsilon} \tag{3}$$

When dealing with limits at infinity, this definition is modified to read for all $\epsilon > 0$, there exists some real number M such that for all $x > M$, $|f(x-l)| < \epsilon$.

This new definition is analogous to the original δ-ϵ definition except instead of x being sufficiently close to a, we are interested in x being sufficiently large, since we're dealing with limits at infinity.

Now imagine all the places where $|f(x) - l| < \epsilon$ could be false. Notice that they only occur when $x \in A_n$. Now imagine some sufficiently large natural number $N \in \mathbb{N}$, such that $\frac{1}{n} < \epsilon$, whatever ϵ is chosen to be.

Let the set

$$U = A_1 \cup A_2 \cup A_3 \cup ... \cup A_{n-1} \tag{4}$$

be the union of sets A_1 to A_{n-1}. Since it is a finite union of finite sets, it is itself finite and must have a maximum.

If we chose $M = \max(U)$, then $\forall x > M$, either

$$x \notin A_n \text{ for any } n \implies |f(x) - 0| = |0 - 0| < \epsilon$$

or

$$x \in A_n. \text{ since } x > M, n \geq N \implies |f(x) - 0| = \frac{1}{n} \leq \frac{1}{N} < \epsilon$$

So in either case $|f(x) - l| < \epsilon$ so by definition (3), the $lim_{x\to\infty} f(x) = 0$

$\square$

2 Exercise 2

Given any $\epsilon > 0$ find a $\delta > 0$ such that for all x satisfying $|x - 3^4| < \delta$ it holds that $|x^{\frac{1}{4}} - 3| < \epsilon$. If we do so, we prove that $\lim_{x\to a} f(x) = l$ with f, l, and a equal to what?

HINT: For every $m \geq 1$ and $a > b > 0$ it holds that $a^{\frac{1}{m}} \leq b^{\frac{1}{m}} + (a-b)^{\frac{1}{m}}$.

2.1 Solution

$$f(x) = x^{\frac{1}{4}}, l = 3, a = 81 \tag{5}$$

So we need to prove that $\lim_{x\to 81} x^{\frac{1}{4}} = 3$

From definition (3) above, we know that $|x - 81| < \delta$ and $|x^{\frac{1}{4}} - 3| < \epsilon$

If $x > 81$ then (from the hint):

$$x^{\frac{1}{4}} \leq 81^{\frac{1}{4}} + (x-81)^{\frac{1}{4}} \implies x^{\frac{1}{4}} - 3 \leq (x-81)^{\frac{1}{4}} \tag{6}$$

If $x < 81$ then (from the hint):

$$81^{\frac{1}{4}} \leq x^{\frac{1}{4}} + (81-x)^{\frac{1}{4}} \implies 3 - x^{\frac{1}{4}} \leq (81-x)^{\frac{1}{4}} \tag{7}$$

From inequalities (6) and (7) we can see that the difference between the function $x^{\frac{1}{4}}$ and the limit 3 is bounded by the 4^{th} root of the difference between x and 81.

So

$$|x^{\frac{1}{4}} - 3| \leq |x - 81|^{\frac{1}{4}} \tag{8}$$

For the $\lim_{x\to 81} x^{\frac{1}{4}} = 3$ to be true, the left hand side of (8) must be $< \epsilon$. If we set the right hand side of (8) to be $< \epsilon$ this property will obviously hold. So

$$|x - 81|^{\frac{1}{4}} < \epsilon \implies |x - 81| < \epsilon^4 \implies \delta = \epsilon^4 \text{ from (3)} . \tag{9}$$

$\therefore \forall \epsilon > 0, \exists \delta = \epsilon^4$ such that if $0 < |x - 81| < \epsilon^4, |x^{\frac{1}{4}} - 3| < \epsilon$, so the $\lim_{x\to 81} x^{\frac{1}{4}} = 3$

$\square$

3 Exercise 3

Using the δ and ϵ definition of a limit, prove that the following limit does not exist:

$$\lim_{x\to 0} f(x) \tag{10}$$

where

$$f(x) := \begin{cases} 1 & \text{if } x = \frac{1}{3^n} \text{ for some } x \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases} \tag{11}$$

3.1 Solution

From (3), to prove a limit does not exist, we must prove the opposite. Namely:

$$\boxed{\text{The function } f \text{ does not approach a limit } l \text{ at } a \text{ if } \exists \epsilon, \text{such that } \forall \delta > 0, \exists x \text{ with } 0 < |x - a| < \delta \text{ and } |f(x - l)| \geq \epsilon} \tag{12}$$

Suppose for contradiction that $\lim_{x\to 0} = l$ for some $l \in \mathbb{R}$ and $\epsilon = \frac{1}{2}$. Then by assumption, $0 < |x| < \delta$ and $|f(x) - l| < \frac{1}{2}$. Now chose some $n \in \mathbb{N}$ sufficiently large such that $\frac{1}{3^n} < \delta$ whatever we chose δ to be. Then there will be some point $x = \frac{1}{3^n}$ where $f(x) = 1$ and some point $x = \frac{1}{4^n} < \frac{1}{3^n} < \delta$ where $f(x) = 0$.

So $|1-l| < \frac{1}{2}$ and $|0-l| < \frac{1}{2}$. Then by the triangle inequality:

$$|1-0| = |(1-l)+(l-0)| \leq |1-l| + |0-l| < \frac{1}{2} + \frac{1}{2} < 1 \implies 1 < 1 \tag{13}$$

Which is a contradiction. $\therefore$ the $\lim_{x\to 0} f(x)$ does not exist.

$\square$

4 Exercise 4

Let $M \in \mathbb{R}$ and $f(x) \leq g(x)$ for every $x \in (-\infty, M]$. Prove, using the δ and ϵ definition of a limit, that if

$$\lim_{x\to-\infty} g(x) = -\infty \text{ then } \lim_{x\to-\infty} f(x) = -\infty \tag{14}$$

4.1 Solution

From (3):

$$\boxed{\text{The function } f \text{ approaches } -\infty \text{ near } -\infty \text{ if: } \forall B \in \mathbb{R}, \exists N < 0, \text{ such that } \forall x < N, f(x) < B} \tag{15}$$

It is given that $\lim_{x\to-\infty} g(x) = -\infty$, so for an arbitrary $B \in \mathbb{R}, \exists N_g$ such that $\forall x < N_g, g(x) < B$.

Let $N_f = \min(N_g, M)$. If $x < N_f$ then since $N_f \leq N_g$ then $g(x) < B$. Also since $N_f \leq M$ the given statement $f(x) \leq g(x)\ \forall x \in (-\infty, M]$ applies. So $f(x) \leq g(x) < B$.

So from (15):

$$\lim_{x\to-\infty} f(x) = -\infty \tag{16}$$

$\square$

5 Exercise 5

5.1

If $f(x) < g(x)$ for all $x \in \mathbb{R}$, and $\lim_{x\to\infty} f(x)$ and $\lim_{x\to\infty} g(x)$ exist and are finite, does it follow that:

$$\lim_{x\to\infty} f(x) < \lim_{x\to\infty} g(x)?$$

5.1.1 Solution

No, it does not follow.

Let $f(x) = 0$ and $g(x) = e^{-x}$. No matter how large x is, $e^{-x} > 0$, so $f(x) < g(x)$ but $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = 0$.

Proof:

From (3), a function f approaches l near ∞ if $\forall\epsilon > 0, \exists M > 0$ such that $\forall x > M$, $|f(x) - l| < \epsilon$. For $f(x) = 0$:

$$|f(x) - l| = |0-0| = 0 < \epsilon, \forall M \in \mathbb{R}. \therefore \lim_{x\to\infty} 0 = 0 \tag{17}$$

And for $g(x) = e^{-x}$:

$$|g(x) - 0| = |e^{-x}| < \epsilon \text{ when } M > \ln\frac{1}{\epsilon} \tag{18}$$

$$\text{Since } |e^{-x}| < \epsilon \implies -x < \ln\epsilon \implies x > \ln\frac{1}{\epsilon}.$$

$\square$

5.2

If $f(x) < g(x)$ for all $x \in [0,1]$, and $\lim_{x\to 0^+} f(x)$ and $\lim_{x\to 0^+} g(x)$ exist and are finite, does it follow that:

$$\lim_{x\to 0^+} f(x) < \lim_{x\to 0^+} g(x)?$$

5.2.1 Solution

No, it does not follow.

$$\text{Let } f(x) = 0 \text{ and } g(x) = \begin{cases} x & x \in (0,1] \\ 1 & x = 0 \end{cases}$$

Then $f(x) < g(x)$ for all $x \in [0,1]$ but $\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} g(x) = 0$.

For $f(x) = 0$, from (3): $|f(x) - 0| = |0| < \epsilon$ for all $\delta > 0$.

For $g(x) = \begin{cases} x & x \in (0,1] \\ 1 & x = 0 \end{cases}$, $|g(x) - 0| < \epsilon \implies |g(x)| < \epsilon$ when $\delta = \min(1, \epsilon)$

Proof:

Let $\epsilon > 0$, choose $\delta = \min(1, \epsilon)$. Then, let x satisfy $0 < |x - 0| < \delta \implies 0 < |x| < \delta$.

Since $0 < |x|$ and $\delta \leq 1$, $g(x) = x$.

So $|g(x) - 0| = |g(x)| = |x| < \delta \leq \epsilon$.

So, from (3), the $\lim_{x\to 0^+} g(x) = 0$.

$\square$